

Q1: Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be differentiable functions satisfying $f'(x) < g'(x)$ for every values of $x \in \mathbb{R}$ and $f(0) = g(0)$.
Prove that $f(x) < g(x)$ for all $x > 0$ and $f(x) > g(x)$ for all $x < 0$.

Solⁿ: Let $h(x) = f(x) - g(x)$
since f and g be differentiable. so, h is differentiable on $\mathbb{R}$

$$h(0) = f(0) - g(0) = 0$$

$$h'(x) = f'(x) - g'(x) < 0 \text{ for all } x \in \mathbb{R}$$

so h is strictly decreasing on $\mathbb{R}$.

Case I: ~~since~~ $x > 0$ and h is decreasing.

$$h(x) < h(0) = 0$$

$$\Rightarrow f(x) - g(x) < 0$$

$$\Rightarrow f(x) < g(x) \quad \forall x \in \mathbb{R}$$

Case II: since $x < 0$, and h is decreasing, so

$$h(x) > h(0) = 0$$

$$\Rightarrow f(x) - g(x) > 0$$

$$\Rightarrow f(x) > g(x) \quad \forall x \in \mathbb{R}$$



Problem: Assume that f has a finite derivative in (a, b) and is continuous on $[a, b]$ with $a \leq f(x) \leq b$ for all x in $[a, b]$ and $|f'(x)| \leq \alpha < 1$ for all x in (a, b) . Prove that f has a unique fixed point in $[a, b]$.

Sol: We want to show that $f(x) = x$.

$$\text{let } g(x) = f(x) - x$$

since f is continuous on $[a, b]$, then g is also continuous on $[a, b]$.

since $f(x) \in [a, b]$, then

$$f(a) \geq a \Rightarrow g(a) = f(a) - a \geq 0$$

$$f(b) \leq b \Rightarrow g(b) = f(b) - b \leq 0$$

if $g(a) = 0 \Rightarrow f(a) = a$ ~~done~~
or $g(b) = 0 \Rightarrow f(b) = b$ ~~done~~ } Nothing to prove.

say $g(a) > 0$, $g(b) < 0$ and g is continuous, then
IVT, there exists a c in (a, b) such that

$$g(c) = 0$$

$$\Rightarrow f(c) - c = 0$$

$$\Rightarrow f(c) = c$$

$$\Rightarrow f(x) = x$$

Uniqueness: let $f(x_1) = x_1$ and $f(x_2) = x_2$ $x_1 \neq x_2$

$$\text{Applying MVT, } f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{x_2 - x_1}{x_2 - x_1} = 1$$

But $|f'(x)| < 1 \quad \forall x \in (a, b)$ which is contradicts.
So, $f(x) = x$ $\rightarrow$

Problem: Use the Mean-Value Theorem to show that if f is differentiable on an interval, and if $|f'(x)| \leq M$ for all values of x in the interval, then $|f(x) - f(y)| \leq M|x - y|$ for all values of x and y in the interval.

Sol: ~~Let~~ we assume that f is differentiable on (x, y) and continuous on $[x, y]$.

By the MVT, $\exists c \in (x, y)$ s.t.

$$f'(c) = \frac{f(y) - f(x)}{y - x}$$

$$\Rightarrow f(y) - f(x) = f'(c)(y - x)$$

$$\Rightarrow |f(y) - f(x)| = |f'(c)| |y - x|$$

since $|f'(x)| \leq M$, so

$$|f(y) - f(x)| \leq M |y - x| \quad \forall x, y \in (x, y) \quad \times$$

Problem: Use the above problem to prove that $|\sin x - \sin y| \leq |x - y|$
 $\forall x, y \in \mathbb{R}$.

Sol: Let say $f(x) = \sin x$

since $\sin x$ is differentiable on $\mathbb{R}$.

so, $f'(x) = \cos x$, and $|\cos x| \leq 1 \quad \forall x \in \mathbb{R}$

Now $|\sin x - \sin y| \leq 1 \cdot |x - y| = |x - y| \quad \forall x, y \in \mathbb{R}$

$\times$

Problem: Let $f: [0, \infty) \rightarrow \mathbb{R}$ be a continuous function such that $f(0) = 0$. If f is differentiable on $(0, \infty)$ and f' is increasing on $(0, \infty)$ then show that g is an increasing function where g defined as $g: (0, \infty) \rightarrow \mathbb{R}$ by $g(x) = \frac{f(x)}{x}$.

Solⁿ: We want to show that $g'(x) \geq 0$ for all $x > 0$.

Here $g(x) = \frac{f(x)}{x}$

$$g'(x) = \frac{x f'(x) - f(x)}{x^2}$$

We want to show $g'(x) \geq 0$

~~so, it is $x f'(x) - f(x) \geq 0 \Rightarrow x f'(x) \geq f(x)$~~

so, it is enough to show that $x f'(x) \geq f(x)$.

Since f is continuous on $[0, x]$ and differentiable on $(0, x)$ and $f(0) = 0$.

By the MVT, there exists some $c \in (0, x)$ s.t.

$$f'(c) = \frac{f(x) - f(0)}{x - 0} = \frac{f(x)}{x}$$

$$f(x) = x f'(c)$$

Since f is increasing and $c < x$, we get:

$$f'(c) \leq f'(x)$$

$$\Rightarrow x f'(c) \leq x f'(x)$$

$$\Rightarrow f(x) \leq x f'(x)$$

$$\Rightarrow x f'(x) - f(x) \geq 0 \Rightarrow g'(x) \geq 0$$

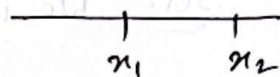
So $g(x)$ is increasing on $(0, \infty)$.

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Problem: Suppose f is continuous on the closed interval $[a, b]$, that f is differentiable on the open interval (a, b) , and that $f'(x) = 0$ for $\forall x \in (a, b)$. Then f is constant on that interval $[a, b]$.

Solⁿ: Let $x_1, x_2 \in [a, b]$ and $x_1 < x_2$

f is continuous on $[x_1, x_2]$ and differentiable on (x_1, x_2) where $[x_1, x_2] \subseteq [a, b]$



By the MVT, $\exists$ some $c \in (x_1, x_2)$ s.t.

$$f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

since $f'(x) = 0 \quad \forall x \in (a, b)$, so

$$f'(c) = 0$$

$$\Rightarrow \frac{f(x_2) - f(x_1)}{x_2 - x_1} = 0$$

$$\Rightarrow f(x_2) = f(x_1)$$

Since x_1 and x_2 are arbitrary in $[a, b]$ which implies $f(x)$ is constant on $[a, b]$.

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page - 123. Exercise:

Problem: 5.17, 5.18

Problem ^(a): Use the Mean-Value Th.^m to show that if f is differentiable on an interval, and if $|f'(x)| \leq M$ for all values of x in the interval, then

$$|f(x) - f(y)| \leq M|x - y|$$

for all values of x and y in the interval.

(b) Use the result in part (a) to show that

$$|\sin x - \sin y| \leq |x - y|.$$

for all real values of x and y .

Problem: (a) Use the M-V-T to show that if f is differentiable on an open interval, and if $|f'(x)| \geq M$ for all values of x in the interval, then

$$|f(x) - f(y)| \geq M|x - y|$$

for all values of ~~and~~ x and y in the interval.

(b) Use the result in part (a) to show that

$$|\tan x - \tan y| \geq |x - y|$$

for all values of x and y in the interval $(-\pi/2, \pi/2)$.

(c) Use (b) show that

$$|\tan x + \tan y| \geq |x + y|, \quad \forall x, y \in (-\pi/2, \pi/2).$$